

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
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Branch	CSIT
Course Outcome	CO4-Floyd-Warshall All-Pairs Shortest Paths

Case Study Topic: Mumbai Dabbawalla Delivery Network — All-Pairs Shortest Path for Traffic Re-routing

Work Done Summary:

In this case study, we analyzed an undirected transport network representing 6 key Mumbai tiffin transfer hubs interconnected by 9 travel-time weighted edges. We initialized the base distance matrix (Sdp^0) mapping direct paths and infinity values, executed the Floyd-Warshall dynamic programming main loop manually for iterations $k=0, 1, 2$ to trace optimization changes via intermediate hops, completed a functional Java program structure with integer overflow guards, and evaluated its $O(V^3)$ computational budget against running multiple iterations of Dijkstra's algorithm.

Implementation Details:

1. Modeled the six-hub delivery network using a six-by-six distance matrix.
2. Initialized direct edge weights, set the diagonal values to zero, and used an infinity constant for missing links.
3. Implemented three nested loops to evaluate each vertex as an intermediate routing hub.
4. Used a minimum-cost recurrence relation to update the optimal travel paths dynamically.
5. Added logical checks to prevent integer overflow when adding infinity values during calculations.
6. Printed the formatted all-pairs shortest time matrix using the station codes.

Result: Screenshots/output

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All-Pairs Shortest Travel Time Matrix (minutes):
      CHC  DOR  RRL  GTR  VKR  TNE
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CHC | 0   10  15  27  20  26
DOR | 10  0   8   18  13  19
RRL | 15  8   0   12  5   11
GTR | 27  18  12  0   7   9
VKR | 20  13  5   7   0   6
TNE | 26  19  11  9   6   0
=== Code Execution Successful ===
```

GitHubLink: https://github.com/Varnika1024/DSA_CASE-STUDY/blob/main/co4_assignment.java